Supplementary Materials for: A Two-Stage Clustered MILP Framework for Efficient and Resilient Waste Collection Routing

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Introduction to the Supplementary Materials

This document accompanies the ICERIE 2027 conference paper (Paper ID 87), referred to below as the *conference paper*. The conference paper had a strict page limit, so it gives a condensed version of the work. The material here comes from the longer case study report on which the conference paper is based (Department of IPE, RUET, April 2026), referred to as the *full paper*. Its purpose is to give readers what they need to inspect and reproduce the work: the complete data table, the cluster assignments, the full mathematical model with its explanations, the unrounded scenario results and the extra sensitivity material.

Nothing in this document is new research. All values were copied from the full paper. Where a value was obtained by simple addition or a percentage from tabulated numbers, it is marked as *derived*. Where either paper does not report something that a reader would need, Section 8 says so instead of filling the gap.

Abbreviations. BDT: Bangladeshi taka; CO_2: carbon dioxide; MILP: mixed-integer linear programming; MSW(M): municipal solid waste (management); MTZ: Miller–Tucker–Zemlin; RCC: Rajshahi City Corporation; STS: secondary transfer station; VRP: vehicle routing problem; CVRPTW: capacitated VRP with time windows.

Supplementary Dataset and STS Information

Data sources

The full paper (Section 4.1) states that data came from 20 STS locations in Rajshahi that cover residential, commercial, institutional and industrial areas. For each STS, latitude and longitude and estimated daily waste generation (tons/day) were recorded. Primary data came from field observations and secondary data from existing records and the literature. STS information is attributed to the assessment of STS in RCC by Ahmad et al. (full-paper ref. [18]); coordinates

and average traffic speeds to Google Maps Platform (ref. [20], accessed 21 March 2026). The full paper gives no further detail on the field survey.

Vehicle fleet and operational parameters

Table S1. *Vehicle fleet specifications (full paper Table 4.1; conference paper Table 1). Capacity in tonnes, fuel cost in BDT per km, wage in BDT per vehicle per day, emission factor in kg CO_2 per km.*

Class	Role (set)	Q_k (t)	C_k^{fuel}	C_k^{wage}	E_k
Type 1 (1.5 T)	Sweeper (K_R)	1.5	18	1200	0.15
Type 2 (3.0 T)	Sweeper (K_R)	3.0	25	1500	0.22
Type 3 (5.0 T)	Shuttle (K_S)	5.0	35	1800	0.35

Table S2. *Operational and environmental parameters (full paper Table 4.2).*

Parameter	Symbol	Value	Source / description (as stated)
Congestion factor	β	1.3–1.5	Based on Rajshahi traffic audits
Maximum working time	$T_{\max}$	8 hours	Standard labour shift
Average speed	v	20 km/h	Typical city dumper speed
Carbon penalty	λ	3.85 BDT/kg CO_2	Cramton and Stoft (2009)

The full paper notes that larger vehicles have higher fuel cost per km, higher wages and higher emission factors, and that this lets the model trade cost against capacity and emissions. In the sensitivity analysis the baseline is $\beta = 1.4$ and $\lambda = 3.85$ (conference paper Section 4.2).

STS data

Table S3. *STS names, coordinates, K-means cluster and daily waste generation (full paper Table 4.3, same values and order). Latitude and longitude in decimal degrees; waste in tons/day. The index n is the label printed on each point in Figure S1; it follows the row order of the original table.*

n	STS name	Latitude	Longitude	Cluster	Waste (t/day)
n	STS name	Latitude	Longitude	Cluster	Waste (t/day)
	Kashiadanga STS	24.3736	88.5778	2	9
2	Adorsho School STS	24.3705	88.5852	2	13
3	Court Walton STS	24.3718	88.5895	2	21
4	New Degree College STS	24.3689	88.5938	2	27
5	Mission Hospital STS	24.3725	88.5962	0	29
6	STS Bondhogate	24.3732	88.5980	0	17
7	Terokhadia STS	24.3768	88.6015	0	13
8	Match Factory STS	24.3785	88.6068	0	24

n	STS name	Latitude	Longitude	Cluster	Waste (t/day)
9	STS near Sericulture	24.3738	88.6035	0	20
10	RMC STS	24.3729	88.6052	0	31
11	TTC STS	24.3702	88.6020	0	14
12	STS Borokuthi	24.3680	88.6005	0	18
13	Tikapara STS	24.3708	88.6115	0	25
14	Kazla STS	24.3695	88.6355	3	27
15	RU STS	24.3678	88.6288	3	28
16	Mansur Ali Park STS	24.3755	88.6102	0	20
17	Nowdapara STS	24.3925	88.6065	0	15
18	Haragram STS	24.3765	88.5616	2	8
19	Sopura STS	24.3660	88.5880	2	13
20	Airport STS	24.4372	88.6160	1	11

The highest generation is at RMC STS (31 t/day) and the lowest at Haragram STS (8 t/day), matching the conference paper. The full paper also names Mission Hospital STS (29 t/day), RU STS (28 t/day) and Airport STS (11 t/day) as examples.

Detailed Clustering Methodology and Results

Method

K-means clustering (MacQueen, 1967; Sinaga and Yang, 2020) partitions the STS using only their geographic coordinates (latitude and longitude). It minimises the within-cluster sum of squared distances, so that nearby STS fall in the same cluster. Each cluster becomes an isolated routing zone with its own sub-fleet, which removes overlapping routes between zones and shrinks the routing problem. The MILP of Section 4 is then solved separately for each cluster. The number of clusters used was $K = 4$ (Cluster 0 to Cluster 3).

Cluster assignments and characteristics

Table S4. *Cluster membership summarised from Table S3. The number of STS and the total nominal daily waste per cluster are derived by addition from Table S3 and are not printed in either paper.*

Cluster	STS (n)	Members (index n from Table S3)	Count	Sum (t/day)
0	5–13, 16, 17	Mission Hospital, Bondhogate, Terokhadia, Match Factory, Sericulture, RMC, TTC, Borokuthi, Tikapara, Mansur Ali Park, Nowdapara	11	226
1	20	Airport	1	11
2	1–4, 18, 19	Kashiadanga, Adorsho School, Court Walton, New Degree College, Haragram, Sopura	6	91

Cluster	STS (n)	Members (index n from Table S3)	Count	Sum (t/day)
3	14, 15	Kazla, RU	2	55
All			20	383

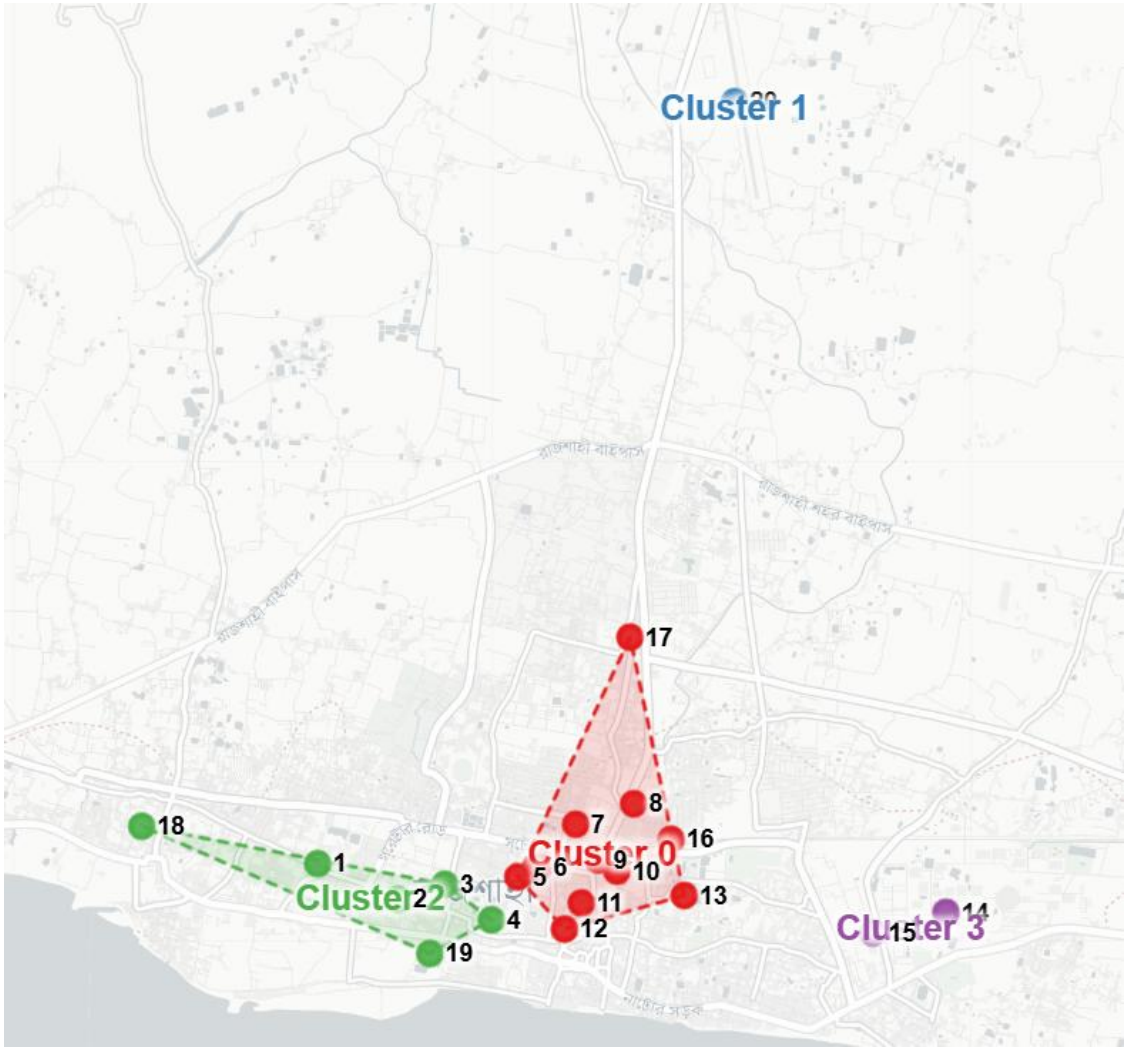


Figure S1. *Clustering of the 20 STS into four clusters by K-means, plotted on a street map of Rajshahi (original Figure 4.1; image as supplied in the full paper). Point labels are the indices n of Table S3. Cluster 0 (red), Cluster 1 (blue, Airport STS), Cluster 2 (green) and Cluster 3 (purple).*

Observation for the authors. The 20 STS in Table S3 add up to 383 t/day, whereas the scenario totals in Table S6 range from 434.7 to 470.6 t/day, and Z_0 is described as the observed baseline

(459.6 t). Neither paper explains the difference (for example whether the scenario totals include a surge term or another waste stream). It is listed in Section 8.

Complete Mathematical Formulation of the Two-Stage MILP Model

The model is written for one cluster. It is a CVRPTW with a heterogeneous fleet of direct-trip shuttles and multi-stop sweepers. The text below follows Section 3.1 of the full paper; equation numbers are those of the full paper. No constraint has been added, removed or altered.

Sets and indices

- $i, j \in \mathcal{N}$: all network nodes in the cluster; node 0 is the central depot (landfill) and $\mathcal{C} = \mathcal{N} \backslash \{0\}$ is the set of STS nodes.
- $k \in K_S$: shuttle vehicles, which do primary bulk extraction by direct depot-to-STS round trips.
- $k \in K_R$: sweeper vehicles, which collect residual waste along multi-stop tours.
- $m \in M = \{1, \ldots, |M|\}$: trip slots; $|M|$ is the maximum number of sequential tours a sweeper can complete in one shift. $|M|$ is set as an upper bound from the cluster's waste volume and the fleet capacity so that the solver does not create trips during the solve.
- $\mathcal{A}$: admissible arcs, $\mathcal{A} = \left\{(i,j) \;\middle|\; \frac{2D_{0i}}{v/\beta} + t_{LU}^{R} \leq T_{\max},\ i \neq j\right\}$.

Parameters

Symbol	Definition	Unit
Q_k	Payload capacity of vehicle k	t
C_k^{fuel}	Fuel cost per km of vehicle k	BDT/km
C_k^{wage}	Fixed daily labour cost of vehicle k	BDT/day
E_k	CO_2 emission factor of vehicle k	kg CO_2/km
λ	Carbon penalty rate	BDT/kg CO_2
$\bar{S}_i$	Nominal waste accumulation at STS i	t
$\hat{S}_i$	Maximum surge demand above the nominal level at STS i	t
Γ	Robustness budget, $\Gamma \in [0,1]$	–
D_{0i}	Road-network distance from the depot to STS i	km
D_{ij}	Road-network distance from node i to node j	km
v	Free-flow travel speed	km/h
β	Congestion multiplier, $\beta \geq 1$; effective speed is v/β	–
t_{LU}	Loading and unloading service time per stop	h
$[E_i, L_i]$	Hard time window at STS i: earliest and latest permitted arrival	h
B_{dur}	Mandatory driver rest per deployed vehicle	h

Symbol	Definition	Unit
$T_{\max}$	Maximum shift duration	h
$\mathcal{M}$	Dynamic big-M constant for the time-continuity linearisation	h

Decision variables

Variable	Meaning
$y_{ik} \in \mathbb{Z}_{\geq 0}$	Number of direct depot-to-STS round trips by shuttle k to STS i per shift
$x_{ijkm} \in \{0,1\}$	1 if sweeper k uses arc (i,j) on trip m, else 0
$q_{ikm} \geq 0$	Residual waste (t) collected by sweeper k at node i on trip m
$T_{ikm} \geq 0$	Arrival time (h from shift start) of sweeper k at node i on trip m
$u_{ikm} \geq 1$	Auxiliary MTZ visit-order variable for sweeper k at node i on trip m
$N_k \in \mathbb{Z}_{\geq 0}$	Number of physical vehicles of type k deployed

Objective function

$$\min Z = Z^{Fuel} + Z^{Wages} + Z^{Carbon} \tag{3.1}$$

$$Z^{Fuel} = \sum_{k \in K_S} \beta \; C_k^{fuel} \left[\sum_{i \in \mathcal{C}} 2 \, D_{0i} \, y_{ik} \right] + \sum_{k \in K_R} \sum_{m \in M} \beta \; C_k^{fuel} \left[\sum_{(i,j) \in \mathcal{A}} D_{ij} \; x_{ijkm} \right] \tag{3.2}$$

$$Z^{Wages} = \sum_{k \in K_S \cup K_R} N_k \; C_k^{wage} \tag{3.3}$$

$$Z^{Carbon} = \sum_{k \in K_S} \lambda \, E_k \left[\sum_{i \in \mathcal{C}} 2 \, D_{0i} \, y_{ik} \right] + \sum_{k \in K_R} \sum_{m \in M} \lambda \, E_k \left[\sum_{(i,j) \in \mathcal{A}} D_{ij} \; x_{ijkm} \right] \tag{3.4}$$

Eq. (3.1) minimises total daily operating cost. Eq. (3.2) is the distance-proportional fuel cost of both fleets, scaled by β. Eq. (3.3) is the fixed daily labour cost per deployed vehicle, independent of distance. Eq. (3.4) converts emissions to cost by applying λ to the distance-based emission factors.

Constraints

$$\sum_{k \in K_S} y_{ik} \geq 1, \qquad \forall i \in \mathcal{C} \qquad (3.5)$$

$$\sum_{k \in K_S} Q_k \; y_{ik} + \sum_{k \in K_R} \sum_{m \in M} q_{ikm} \geq \bar{S}_i + \Gamma \, \hat{S}_i, \qquad \forall i \in \mathcal{C} \qquad (3.6)$$

$$\sum_{j \in \mathcal{N}} x_{ijkm} = \sum_{j \in \mathcal{N}} x_{jikm}, \qquad \forall i \in \mathcal{N}, \; k \in K_R, \; m \in M \qquad (3.7)$$

$$\sum_{j \in \mathcal{C}} x_{0jkm} \leq 1, \qquad \forall k \in K_R, \; m \in M \qquad (3.8a)$$

$$\sum_{j \in \mathcal{N}} x_{jikm} \leq 1, \qquad \forall i \in \mathcal{C}, \; k \in K_R, \; m \in M \qquad (3.8b)$$

$$q_{ikm} \leq Q_k \sum_{j \in \mathcal{N}} x_{ijkm}, \qquad \forall i \in \mathcal{C}, \; k \in K_R, \; m \in M \qquad (3.9)$$

$$\sum_{i \in \mathcal{C}} q_{ikm} \leq Q_k \sum_{j \in \mathcal{C}} x_{0jkm}, \qquad \forall k \in K_R, \; m \in M \qquad (3.10)$$

$$u_{ikm} - u_{jkm} + |\mathcal{C}| \, x_{ijkm} \leq |\mathcal{C}| - 1, \qquad \forall i, j \in \mathcal{C}, \; i \neq j, \; k \in K_R, \; m \in M \qquad (3.11)$$

$$E_i \sum_{j \in \mathcal{N}} x_{ijkm} \leq T_{ikm} \leq L_i \sum_{j \in \mathcal{N}} x_{ijkm}, \qquad \forall i \in \mathcal{C}, \; k \in K_R, \; m \in M \qquad (3.12)$$

$$T_{jkm} \geq T_{ikm} + \frac{D_{ij}}{v/\beta} + t_{LU} - \mathcal{M}\left(1 - x_{ijkm}\right), \qquad \forall (i,j) \in \mathcal{A}, \; j \neq 0, \; k \in K_R, \; m \in M \qquad (3.13)$$

$$\mathcal{M} = T_{\max} + \max_{(i,j) \in \mathcal{A}} \left[\frac{D_{ij}}{v/\beta} + t_{LU}\right] \qquad (3.14)$$

$$\sum_{m \in M} \sum_{(i,j) \in \mathcal{A}} \left[\frac{D_{ij}}{v/\beta} + t_{LU}\right] x_{ijkm} + B_{dur} \leq T_{\max}, \qquad \forall k \in K_R \qquad (3.15a)$$

$$\sum_{i \in \mathcal{C}} y_{ik} \left[\frac{2D_{0i}}{v/\beta} + t_{LU}\right] + B_{dur} \leq T_{\max}, \qquad \forall k \in K_S \qquad (3.15b)$$

$$N_k \geq \sum_{j \in \mathcal{C}} x_{0jkm}, \qquad \forall k \in K_R, \; m \in M \qquad (3.16)$$

$$N_k \geq 1/5 \sum_{i \in \mathcal{C}} y_{ik}, \qquad \forall k \in K_S \qquad (3.17)$$

Interpretation of the constraints

- (3.5) Every STS gets at least one shuttle round trip per shift, which guarantees a minimum level of bulk extraction.

- (3.6) Robust demand satisfaction: combined shuttle and sweeper service must meet or exceed worst-case demand, $\bar{S}_i + \Gamma \hat{S}_i$.
- (3.7) Flow balance: arcs entering a node equal arcs leaving it, for each vehicle, trip and node.
- (3.8a) At most one depot departure per sweeper and trip slot, which prevents artificial route fragmentation. (3.8b) At most one visit per STS in a trip.
- (3.9) Waste is collected at a node only if the node is visited. (3.10) Waste collected on a trip cannot exceed vehicle capacity; unused trips are switched off automatically.
- (3.11) MTZ subtour elimination (Desrochers and Laporte, 1991) through a consistent visiting order.
- (3.12) Hard time windows apply only at visited nodes. (3.13) Arrival times propagate along active arcs with travel time and service time. (3.14) A tight big-M value improves numerical stability and the linear relaxation.
- (3.15a) Total sweeper operating time (travel, service, mandatory rest) stays within the shift. (3.15b) The analogous limit for shuttles uses cumulative round-trip service time.
- (3.16) Lower bound on deployed sweepers from activated trip slots. (3.17) Lower bound on deployed shuttles, reflecting at most five feasible round trips per shift.

Assumptions and boundary conditions

1. All STS are fully operational over the planning horizon, with no disruptions or capacity limits (conference paper A1).
2. All vehicles have identical carbon emission rates regardless of age, type or condition. The conference paper restates this as: vehicles of the same class have identical emission rates (A2).
3. Fuel efficiency is uniform across the fleet (conference paper: identical within a class).
4. Speed is constant on all arcs within a scenario (A3).
5. The depot has an unlimited supply of shuttles, sweepers and drivers, so any daily demand can be met (A4).

Solution approach

The full paper follows a cluster-first, route-second strategy. Stage 1 applies K-means to STS coordinates. Stage 2 solves the MILP separately for each cluster with Gurobi, treating it as a CVRPTW: shuttles take bulk waste and sweepers collect residual waste across several STS. The robustness of the plan is then tested with scenario-based stochastic simulation (Z_0 to Z_{10}) and sensitivity analysis. The overall flowchart and the single-cluster routing sketch appear as Figures S3 and S4 in Section [sec:figs].

Table S5. *Equation numbering in the full paper and in the conference paper.*

Full paper	Conference paper
(3.1), (3.2), (3.4), (3.3)	(1), (3), (4), (2)
(3.5), (3.6)	(5), (6)
(3.7)	(7)
(3.8a), (3.8b)	(8), combined
(3.9), (3.10)	(9), combined
(3.11)	(10)
(3.12), (3.13), (3.14)	(11), (12), (13)
(3.15a), (3.15b)	(14), (15)
(3.16), (3.17)	(16), (17)

The conference paper writes (3.8a) and (3.8b) as one line (8) and (3.9) and (3.10) as one line (9). It also writes the quantifier of (3.9)/(3.10) as $\forall i \in \mathcal{C}$ for both, whereas in the full paper (3.10) is stated per (k, m) only. The two forms are logically the same, because (3.10) does not depend on i.

Detailed Optimization Results

Traditional versus optimized cost by scenario

The full paper compares a traditional system with the optimized solution from Gurobi in eleven scenarios. Z_0 is the baseline (observed) case and Z_1 to Z_{10} are simulated stochastic demand scenarios (legend of Figure 4.2). The traditional cost is described there as an estimate for fixed-route, direct depot-to-STS dispatch with no route optimization and no demand uncertainty. The optimized cost is described as the cluster-wise MILP with robust demand buffering ($\Gamma = 0.2$), time-window constraints and multi-shift scheduling.

Table S6. *Daily collection cost of traditional dispatch and the optimized MILP solution (full paper Table 4.4, full precision). Waste is the total daily waste in tons; costs in BDT/day. The conference paper (Table 2) shows the same values rounded to whole BDT.*

Scenario Z_n	Waste (t)	Traditional (BDT)	Optimized (BDT)	Saved (BDT)	Reduction (%)
0	459.60	119,723.35	105,223.67	14,499.68	12.11
1	441.69	117,160.22	96,636.83	20,523.39	17.52
2	470.60	119,343.62	104,943.62	14,400.00	12.07
3	440.88	112,451.35	98,531.63	13,919.73	12.38
4	441.41	115,601.46	98,114.24	17,487.22	15.13
5	452.90	121,108.08	101,720.54	19,387.54	16.01
6	470.52	121,422.26	104,902.52	16,519.74	13.61
7	463.63	117,123.26	104,523.26	12,600.00	10.76

Scenario Z_n	Waste (t)	Traditional (BDT)	Optimized (BDT)	Saved (BDT)	Reduction (%)
8	452.16	113,738.94	96,815.05	16,923.90	14.88
9	434.68	115,911.26	100,471.30	15,439.95	13.32
10	439.72	120,810.39	100,616.72	20,193.67	16.72

Checks. Reductions were recomputed as (Traditional − Optimized)/Traditional and agree with the printed percentages to two decimals. The range is 10.76% (Z_7) to 17.52% (Z_1) and 12,600 to 20,523.39 BDT/day, as in the conference paper. The unweighted mean of the eleven percentages is 14.04% (derived); the conference paper reports 14.05%, which equals the pooled reduction of total saved over total traditional cost (derived). The difference is rounding only.

Route allocation and collection information

Neither paper reports the optimized routes, the number of vehicles deployed per class or cluster (N_k), trip counts per sweeper, route distances, or solver output. The full paper states only that the model turns the conventional radial dispatch into a multi-trip routing network and reduces fleet size and redundant travel. These results are not available in the uploaded documents and are requested in Section 8.

Supplementary Economic Analysis

Economic comparison

Figure S2 shows Table S6 as a bar chart. The conference paper conveys the same information in Table 2 and does not include the figure. The bar labels of Z_9 and Z_{10} are partly hidden by the figure's text box in the original image; the numerical values for those scenarios are in Table S6.

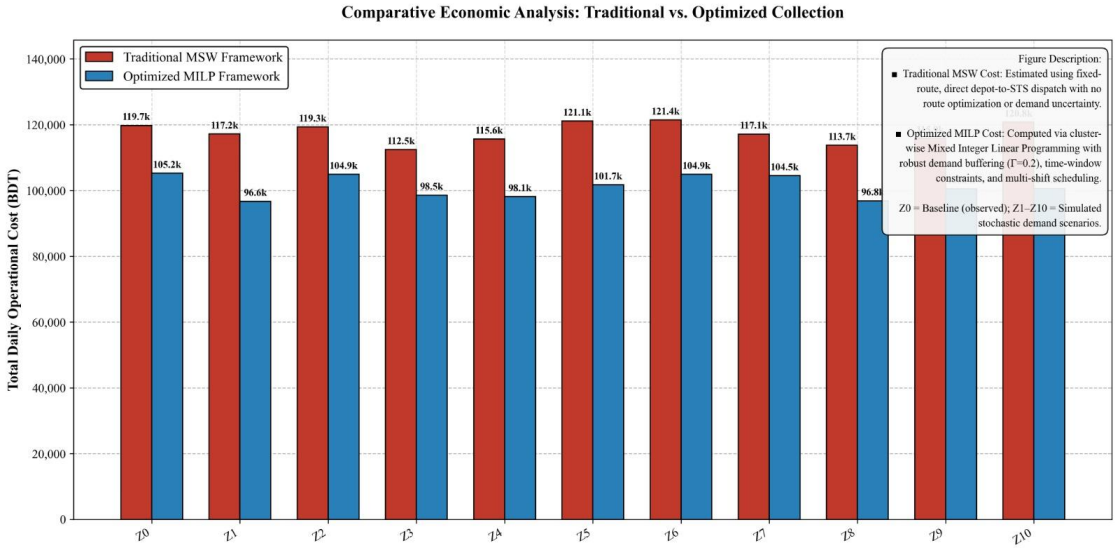


Figure S2. *Comparative economic analysis: traditional versus optimized collection (original Figure 4.2). Total daily operational cost in BDT for scenarios Z_0 to Z_{10}.*

Sensitivity analysis

The key parameters are varied one at a time around the baseline of the conference paper ($\lambda = 3.85$ BDT/kg CO_2, fuel price 100%, $\beta = 1.4$). Table S7 gives the values of conference-paper Table 3 at the precision printed there, with percentage changes derived by division. The corresponding plots are Fig. 2a–c of the conference paper.

Table S7. *Total daily cost Z (BDT) for variation of the carbon tax λ, the fuel price level and the traffic congestion factor β (conference paper Table 3). The change columns are derived: for λ relative to $\lambda = 0$; for fuel price relative to the 100% row (98,806.25); for β relative to $\beta = 1.4$ (98,806.251).*

λ	Z	Change (%)	Fuel price	Z	Change (%)	β	Z	Change (%)
0	95,406.160	0	80%	86,581.190	−12.37	1.0	79,541.878	−19.50
2	96,249.323	+0.88	90%	92,693.720	−6.19	1.2	88,274.064	−10.66
3.85 (baseline)	97,036.715	+1.71	100% (baseline)	98,806.250	0	1.4 (baseline)	98,806.251	0
6	97,954.351	+2.67	110%	104,918.781	+6.19	1.6	107,471.296	+8.77
8	98,811.913	+3.57	120%	111,031.312	+12.37	1.8	117,914.829	+19.34

Across the tested ranges, raising the carbon tax from 0 to 8 BDT/kg CO_2 raises the cost by 3.57%, raising the fuel price from 80% to 120% raises it by 28.24%, and raising β from 1.0 to 1.8 raises it by 48.24% (conference paper, Fig. 2a–c). These range changes match the values in Table S7.

Two-way sensitivity of fuel price and congestion

The conference paper (Fig. 2d and Section 4.2) reports the following for the combined analysis: the cost runs from 70.8k BDT ($\beta = 1.0$, lowest fuel price level) to 134.6k BDT ($\beta = 1.6$, highest fuel price level), which is 28.3% below and 36.2% above the 98.8k BDT baseline. The highest fuel price level alone ($\beta = 1.4$) gives 123.0k BDT (+24.5%), $\beta = 1.6$ alone gives 107.5k BDT (+8.8%), and the two together give 134.6k BDT (+36.2%). Because the two factors compound at the upper end, the paper recommends dynamic re-optimization over static daily budgeting.

Interpretation

Congestion and fuel price are the dominant operational risk factors: congestion extends travel time, raises fuel consumption and lowers vehicle utilization, and fuel price acts directly on transport cost. The carbon tax adds only about 3.6% over the tested range, so it mainly promotes

environmentally sustainable transport decisions, while traffic management and fuel optimization offer larger economic benefits (conference paper Section 4.3).

Additional Figures and Visualizations

The sensitivity plots are Fig. 2 of the conference paper and are not repeated here.

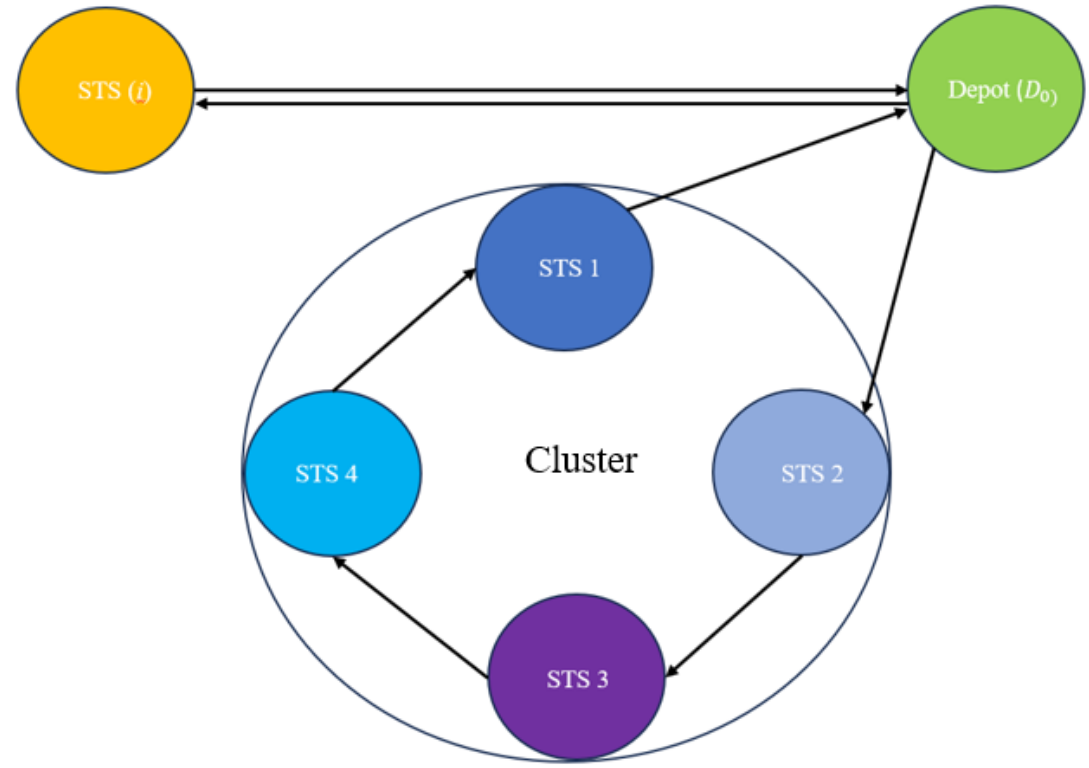


Figure S3. *Vehicle routing in a single cluster (original Figure 3.1): a sweeper leaves the depot, visits several STS in sequence and returns to the depot. This is the same content as Fig. 1b of the conference paper.*

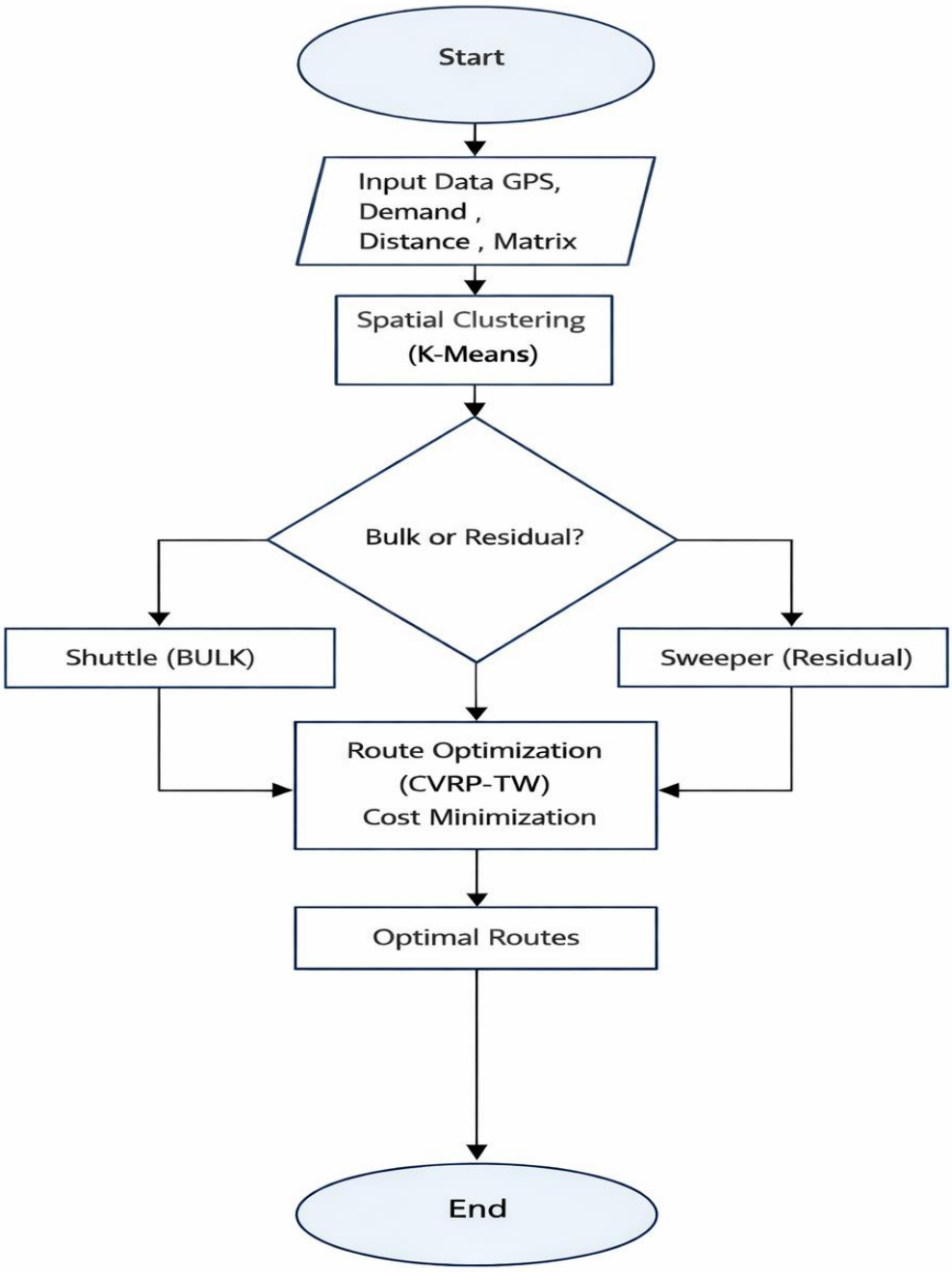


Figure S4. *Flowchart of the overall process (original Figure 3.2). The conference paper shows the workflow in Fig. 1a.*

Reproducibility Notes and Data Availability

Data and code availability

The numerical data in this supplement are in Tables S1–S7. The STS dataset (Table S3) and the cost tables (S6 and S7) can be released as CSV files alongside this PDF. The Gurobi model and K-means scripts were not among the uploaded documents, so no code is described here.

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